

4.1.1. Pandas - series creation and manipulation

03:16

Write a Python program that takes a list of numbers from the user, creates a Pandas series from it, and then calculates the mean of even and odd numbers separately using the `groupby` and `mean()` operations.

Hint: You can group the Series based on whether each number is even or odd.

Input Format:

- The user should enter a list of numbers separated by space when prompted.

Output Format:

- The program should display the mean of even and odd numbers separately.
- Each mean value should be displayed with a label indicating whether it corresponds to even or odd numbers.

Refer to the sample test cases for better understanding regarding input and output format.

Code:

```
import pandas as pd

# Take inputs from the user to create a list of numbers
numbers = list(map(int, input().split()))

s=pd.Series(numbers)

# Create a Pandas series from the list of numbers


# Grouping by even and odd numbers and calculating the mean
grouped = s.groupby(s%2==0).mean()

# Display the mean of even and odd numbers with labels
```

```
grouped.index = ['Even' if is_even else 'Odd' for is_even in grouped.index]
```

```
print("Mean of even and odd numbers:")
```

```
print(grouped)
```

Explorer

Average time
0.043 s
43.50 ms

Maximum time
0.127 s
127.00 ms

✓ 3 out of 3 shown test case(s) passed

✓ 3 out of 3 hidden test case(s) passed

✓ Test case 1 127 ms Debug

Expected output

1 2 3 4 5 6 7 8 9 10

Mean of even and odd numbers:

Odd ... 5.0

Even ... 6.0

dtype: float64

Actual output

1 2 3 4 5 6 7 8 9 10

Mean of even and odd numbers:

Odd ... 5.0

Even ... 6.0

dtype: float64

✓ Test case 2 29 ms Debug

Expected output

4 4 4 6 6 2 2 2 10 12 14 16

Mean of even and odd numbers:

Even ... 6.833333

dtype: float64

Actual output

4 4 4 6 6 2 2 2 10 12 14 16

Mean of even and odd numbers:

Even ... 6.833333

dtype: float64

✓ Test case 3 31 ms Debug

Expected output

1 5 7 9 11 13 15 17 111 21 25

Mean of even and odd numbers:

Odd ... 21.363636

dtype: float64

Actual output

1 5 7 9 11 13 15 17 111 21 25

Mean of even and odd numbers:

Odd ... 21.363636

dtype: float64